AMD DEVELOPER HACKATHON ACT III | TRACK 1 · INTELLIGENT INDUSTRY | ACT I — 1ST PLACE AI AGENTS | OPEN MODELS ON AMD · EVOLUS

FACTORY EVIDENCE

Supplier emissions documents in any language → **verified CBAM savings** for EU steel importers. **The model reads, rules decide, a person approves.**

€636 → €63

CBAM PER TONNE · SLAB · 2026 · MODELED

−66%

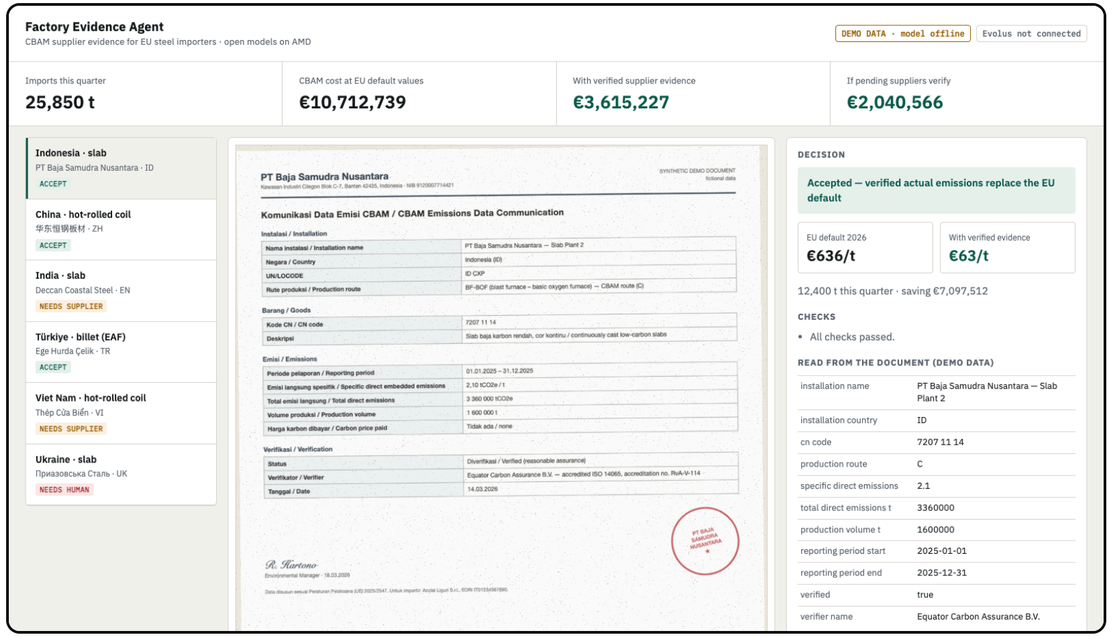
CBAM COST · DEMO QUARTER · MODELED

123

COUNTRIES · OFFICIAL EC TABLES

34 / 34

TESTS · DOCKER · MIT



THE PROBLEM · ONE PERSONA

Every tonne without evidence pays the EU default.

The compliance manager at an Italian steel importer

Since 1 Jan 2026 CBAM charges every imported tonne of steel, aluminium, cement, fertiliser	**definitive period**
No verified supplier data → **EU default value + mark-up**	**+10% → +20% → +30%**
2,400+ authorised CBAM declarants in Italy	**top-5 EU importer**
Italian buyers call the defaults **"unworkable"**; slab purchases stalled	**Eurometal, 2026**
The evidence exists — in supplier scans in **Chinese, Turkish, Indonesian, Ukrainian, Vietnamese**	**manual chase**

MODELED · OFFICIAL EC TABLES · SLAB CN 7207 11 14 · 2026

ORIGIN	EU DEFAULT	ROUTE
Indonesia	**€636 / t**	BF-BOF (C)
India	**€277 / t**	BF-BOF (C)
China	**€177 / t**	BF-BOF (C)
Ukraine	**€108 / t**	BF-BOF (C)
Türkiye	**€100 / t**	BF-BOF (C)

EC default values 06.08.2026 (Annex I 2025/2621 as amended by 2026/1740), benchmarks 06.02.2026 (2025/2620, col. B), CBAM factor 0.975, certificate price €82.32 (2026). Italian context: Bird & Bird, Andersen, Eurometal 2026.

THE 10-SECOND PICTURE

A scan goes in. A decision with euros comes out.



PT Baja Samudra Nusantara
Kawasan Industri Cilegon Blok C-7, Banten 42435, Indonesia · NIB 9120007714421

SYNTHETIC DEMO DOCUMENT
fictional data

Komunikasi Data Emisi CBAM / CBAM Emissions Data Communication

Instalasi / Installation

Nama instalasi / Installation name	PT Baja Samudra Nusantara — Slab Plant 2
Negara / Country	Indonesia (ID)
UN/LOCODE	ID CXP
Rute produksi / Production route	BF-BOF (blast furnace – basic oxygen furnace) — CBAM route (C)

Barang / Goods

Kode CN / CN code	7207 11 14
Deskripsi	Slab baja karbon rendah, cor kontinu / continuously cast low-carbon slabs

Emisi / Emissions

Periode pelaporan / Reporting period	01.01.2025 – 31.12.2025
Emisi langsung spesifik / Specific direct embedded emissions	2,10 tCO2e / t
Total emisi langsung / Total direct emissions	3 360 000 tCO2e
Volume produksi / Production volume	1 600 000 t
Harga karbon dibayar / Carbon price paid	Tidak ada / none

Verifikasi / Verification

Status	Diverifikasi / Verified (reasonable assurance)
Verifikator / Verifier	Equator Carbon Assurance B.V. — accredited ISO 14065, accreditation no. RvA-V-114
Tanggal / Date	14.03.2026

PT BAJA SAMUDRA NUSANTARA ★

R. Hartono
Environmental Manager · 18.03.2026

Data disusun sesuai Peraturan Pelaksana (UE) 2025/2547. Untuk importir: Acciai Liguri S.r.l., EORI IT01234567890.

Synthetic supplier document (fictional company) · values as extracted by the app

ACCEPT — verified actual emissions replace the EU default

EU DEFAULT 2026
€636/t

WITH VERIFIED EVIDENCE
€63/t

READ FROM THE DOCUMENT · EVERY VALUE WITH ITS SOURCE LINE

installation name	**PT Baja Samudra Nusantara – Slab Plant 2**
installation country	**ID**
cn code	**7207 11 14**
good description	**Slab baja karbon rendah, cor kontinu / continuously cast low-carbon slabs**
production route	**C**
specific direct emissions	**2.1**
total direct emissions t	**3360000**

ACCEPT
Indonesia · Türkiye · China

NEEDS SUPPLIER
India unverified · Viet Nam missing CN, period

NEEDS HUMAN
Ukraine 0.35 t/t vs 1.364 benchmark

HOW IT WORKS

The model reads. Rules decide. A person approves.

1 · Read

Open vision model served on AMD (Token Factory, OpenAI-compatible) reads scans in any language and returns each field **with the verbatim line** it came from. No quote, no value.

2 · Check

Deterministic rules, not the model: accredited verification, plausibility vs the EU benchmark of the route, arithmetic (total ÷ volume), reporting period, origin.

3 · Decide

ACCEPT · NEEDS SUPPLIER · NEEDS HUMAN. Nothing is auto-accepted when numbers don't add up — uncertainty is shown, not hidden.

4 · Act

An **Evolus agent** on the same AMD model writes the request **in the supplier's language**, from validator facts and a CBAM rules library. Evolus **anonymize** cleans the auditor file. **A person approves**.

5 · Count

€ per supplier and per quarter: EU default vs verified evidence vs potential if pending suppliers verify. Official EC tables, 123 countries.

Module B · Transizione 5.0: a baseline regression on production, weather and hour-of-week measures an efficiency retrofit with a 95% interval and drafts the note for the certified energy manager. No language model in the numbers.

MEASURED, NOT PROMISED

What we measured this week

WHAT	RESULT	STATUS
Model on AMD (AMD Token Factory), the same model behind the Evolus agent	**Gemma-4-31B-IT**	MEASURED
Field accuracy vs truth · 6 docs · 6 languages (last run · all runs)	**65/66** (98.5%) · range 92.4–98.5% over 5 runs	MEASURED
Decisions correct · false accepts (all runs)	**6/6** last run · **0** false accepts in 30 documents	MEASURED
Model misread a CN code; the rules caught it	**3** of 30 reads · never auto-accepted · 1 fixed by a rules-guided re-read	MEASURED
Latency on the shared AMD endpoint (4 documents in parallel)	median **19–55 s** per page · 6 documents in 54–109 s	MEASURED
Evolus agent on the space's own AMD model + AI Tool “anonymize”	**3/3** supplier requests written · **9/9** Evolus API calls OK	MEASURED
CBAM calculator vs verified model (18 country × good × year cases)	**18 / 18** within €1 / t	MEASURED
Official EC tables parsed	**123** countries · **10,910** default rows · **570** benchmark codes	MEASURED
Validator rules on known cases	accept / unverified / implausible / arithmetic / stale / origin — all caught	MEASURED
Energy baseline on synthetic meter data (true saving 7.20%)	**7.19%** measured · 95% interval 6.92–7.47% · R^2 0.991	MEASURED
Test suite	**34 / 34** passing · Docker · MIT	MEASURED

Supplier documents and meter data are synthetic (fictional companies). Numbers re-generated from the app by **scripts/build_deck.py**; AMD inference panel in the live demo shows per-call latency and tokens.

THE MONEY · MODELED · 2026 PRICES

One quarter of imports: €10,712,739 → €3,615,227

SUPPLIER DOCUMENT	DECISION	TONNES	AT EU DEFAULT	WITH EVIDENCE	IF SUPPLIERS VERIFY
Indonesia · slab	ACCEPT	12400	€7,883,548	€786,036	€786,036
China · hot-rolled coil	ACCEPT	—	—	—	—
India · slab	NEEDS SUPPLIER	8150	€2,259,017	€2,259,017	€684,356
Türkiye · billet (EAF)	ACCEPT	—	—	—	—
Viet Nam · coil	NEEDS SUPPLIER	—	—	—	—
Ukraine · slab	NEEDS HUMAN	5300	€570,174	€570,174	€570,174
Quarter total		**25,850**	**€10,712,739**	**€3,615,227**	**€2,040,566**

−66%

CBAM COST WITH THE EVIDENCE ALREADY ACCEPTED

€572k

LESS CBAM PER 1,000 T OF INDONESIAN SLAB

Modeled from official EC default values and benchmarks; certificate price €82.32 (2026, EC). Tonnages from the synthetic import register; suppliers without a register line show "—".

WHO PAYS · HOW IT REACHES THEM

Priced at a fraction of the saving.

Buyer

Authorised CBAM declarants — steel, aluminium, cement and fertiliser importers. **2,400+ in Italy** alone. Pain is quarterly and grows every year with the mark-up.

Model (proposed)

SaaS plan per declarant + a fee per verified tonne. Example: 1,000 t of Indonesian slab saves ~€572k; a fee of even €1/t is <0.2% of that saving.

Channel

Customs brokers, CBAM consultants and audit firms — each serves hundreds of importers. **Evolus** as the delivery layer for Italian SMEs: workflows, email, Teams, phone.

Next

Same document intake for **EUDR** (deforestation, Dec 2026) and the **battery Digital Product Passport** (Feb 2027). Module B: Transizione 5.0 energy evidence for manufacturers.

WHY IT MATTERS TO EACH SIDE · AND WHAT WE DON'T CLAIM

Everyone gets something real.

AMD
Confidential trade documents read by open models on AMD infrastructure — a workload that should not go to a closed API.

Open models
Multilingual document reading where an open model is the right tool, with every value traceable to its source line.

Evolus
A real, recurring compliance process on its building blocks, starting where Italian SMEs need it most.

Importers & advisors
€ saved per tonne per supplier, and a cited evidence file instead of chasing emails.

What we don't claim

- Documents and meter data in the demo are **synthetic**
- Prices after 2026 are **scenarios**, not forecasts
- The declaration stays with the **authorised declarant** and the **accredited verifier**
- No autonomous compliance decision — **a person approves**
- Quotes are transcribed by the model: in 1 of 66 fields the quote had a typo (value correct)
- Shared AMD endpoint, not a dedicated GPU: 19–55 s per page
- No customer or partner endorsement implied

BUILT THIS WEEK · SOLO

Sardor Razikov · ships and measures.

AMD Developer Hackathon ACT I — 1st place, AI Agents (RepoMind, measured on MI300X)

Kaggle SPR 2026 — **#7 / 371** (top 1.9%)

ML engineer & founder, Tashkent, Uzbekistan

Built in 4 days: EC table parser · calculator · validator · vision extraction · letters · Evolus client · energy module · UI · 34 tests

The ask

Three Italian importers or CBAM advisors to pilot on **real** supplier documents — data processed by open models on AMD, every number traceable.

Repo · github.com/SRKRZ23/factory-evidence
Demo · factory-evidence.netlify.app